

Polish Demographic Structure by Voivodeship, 1995-2024

1 Aim and data source

The aim of this project is to compare demographic ageing across Polish voivodeships and to check whether some regions can be described as relatively older or younger than others. The analysis uses population counts from Bank Danych Lokalnych, published by GUS (Polish Central Statistical Office): <https://bdl.stat.gov.pl/bdl/dane/podgrup/tablica>. The dataset contains population by region, age group, sex, and year.

Because voivodeships differ strongly in total population size, raw counts were transformed into shares of each region's total population. The main variable is therefore the percentage of a voivodeship's population belonging to a selected age group. This makes the regions comparable and is more appropriate for identifying older and younger demographic structures.

2 Data preparation and age categories

The original table was reshaped from a wide format into a tidy long format. Column names containing age group, sex, year, and unit were split into separate variables. The analysis uses only total population for both sexes combined and the years 1995-2024.

Some detailed age groups were not available for the full time interval. In particular, groups 70-74, 75-79, 80-84, and 85+ started in 2002, while 0-14 started in 2013. These overlapping or incomplete groups were removed. The remaining groups were aggregated into six broad life-stage categories:

- Early childhood: 0-9,
- Adolescence: 10-19,
- Young adulthood: 20-29,
- Middle adulthood: 30-44,
- Late adulthood: 45-64,
- Senior population: 65+.

The senior population share is used as the main indicator of how "old" a region is. A low senior share, especially together with relatively larger young age groups, is interpreted as a younger demographic structure. This is a descriptive comparison rather than a formal inferential test.

3 Descriptive statistics for 2024

Table 1 summarizes the 2024 distribution of age-group shares across the 16 voivodeships. All values are percentages of total regional population. The required descriptive measures are included: mean, median, standard deviation, first and third quartiles, range, and interquartile range.

Table 1: Descriptive statistics of age-group shares across voivodeships in 2024.

Age category	Mean	Median	SD	Q1	Q3	Range	IQR
Early childhood (0-9)	9.08	8.81	0.68	8.64	9.55	2.04	0.92
Adolescence (10-19)	10.62	10.79	0.45	10.25	10.95	1.49	0.70
Young adulthood (20-29)	10.02	10.03	0.47	9.65	10.42	1.55	0.77
Middle adulthood (30-44)	22.60	22.40	0.85	22.24	23.35	2.73	1.11
Late adulthood (45-64)	26.81	27.03	0.86	26.06	27.23	3.25	1.17
Senior population (65+)	20.87	20.95	1.36	19.59	21.71	4.41	2.11

The largest average category in 2024 is late adulthood (45-64), at 26.81%, followed by middle adulthood (30-44), at 22.60%, and seniors (65+), at 20.87%. The senior group has the largest standard deviation (1.36 percentage points) and the largest range (4.41 percentage points), which means that regional differences are especially visible for this age group.

3.1 Outlier check

Possible outliers were checked using the standard interquartile rule: values below $Q1 - 1.5 \times IQR$ or above $Q3 + 1.5 \times IQR$ are treated as outliers. For the senior population in 2024, $Q1 = 19.59$, $Q3 = 21.71$, and $IQR = 2.11$. This gives a lower bound of 16.43% and an upper bound of 24.88%. The observed senior shares range from about 18.8% to 23.2%, so no voivodeship is a formal IQR outlier. Nevertheless, the highest and lowest regions are important extremes for interpretation.

4 Graphical analysis

The histogram shows how age-group shares are distributed across the 16 voivodeships in 2024. The distributions show that some age groups vary more across regions than others. The senior population share has a relatively wide spread, which supports the idea that ageing is not equally advanced in every voivodeship.

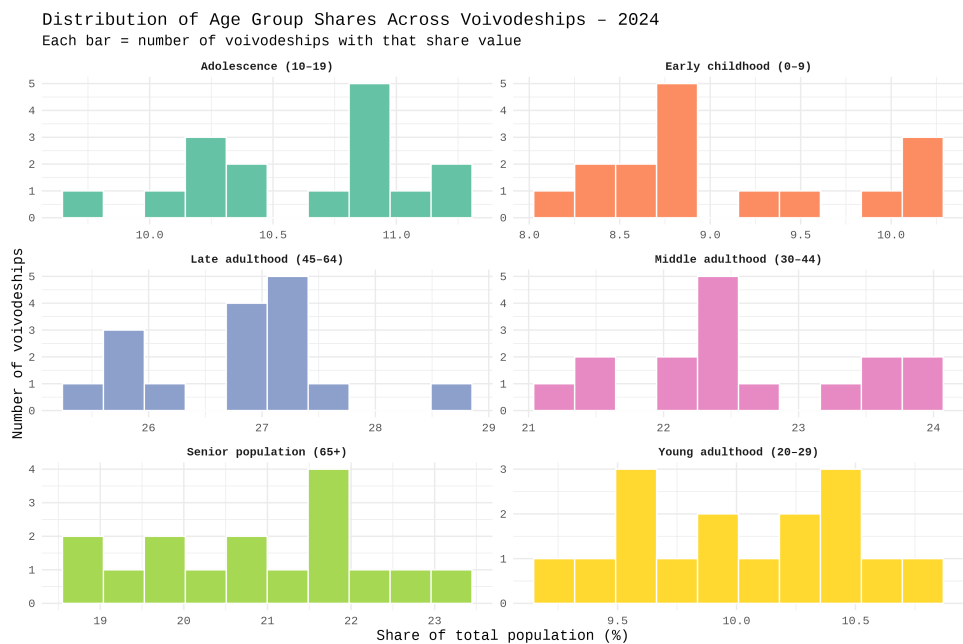


Figure 1: Distribution of age-group shares across voivodeships in 2024. It shows how concentrated or spread out the 16 regional values are within each age group.

Figure 2 focuses on three groups directly connected with the project question: early childhood, young adulthood, and seniors. It shows a strong increase in the senior share in every voivodeship, while early childhood generally declines or stays low. Young adulthood rises until the mid-2000s in many regions and then decreases.

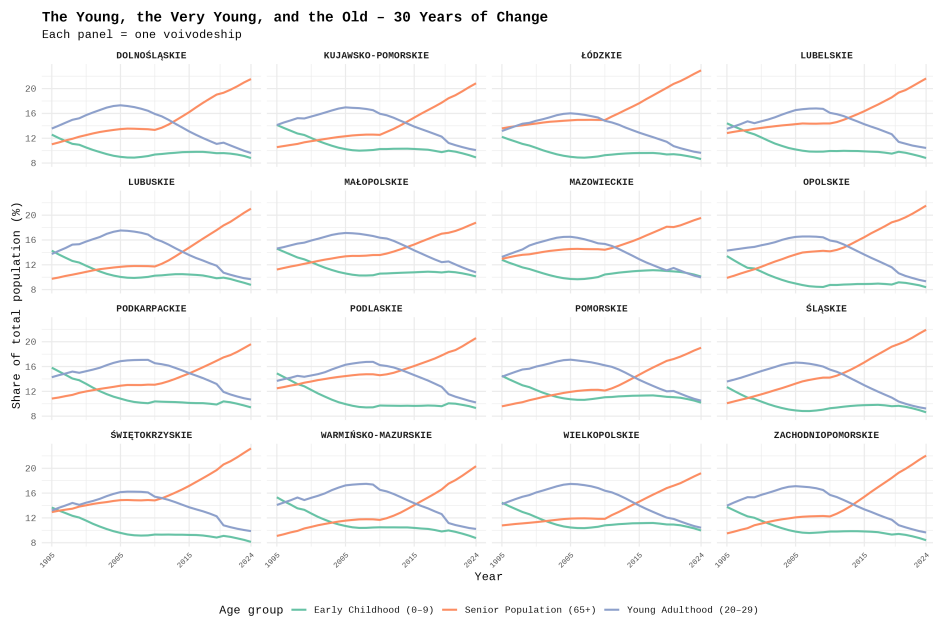


Figure 2: Changes in early childhood, young adulthood, and senior population shares by voivodeship, 1995-2024.

Figure 3 ranks voivodeships by senior share in 2024. A ranking plot is useful because it directly answers the main question: which regions are the oldest according to the senior-share measure. Świętokrzyskie has the highest senior share, at about 23.2%, followed by Łódzkie at 22.9% and Za-

chodniopomorskie at 22.0%. The lowest senior shares are found in Małopolskie, Pomorskie, and Wielkopolskie. These regions can be interpreted as relatively younger under the chosen definition.

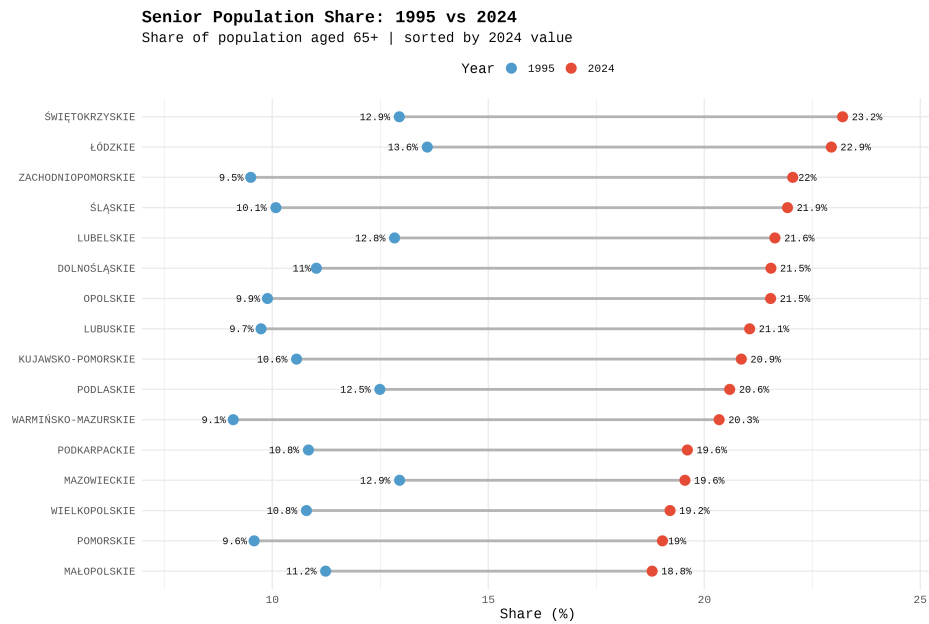


Figure 3: Voivodeships ranked by the share of senior population (65+) in 2024 compared to 1995.

Figure 4 shows the demographic changes across regions in the beginning and end of the measurements. It is very clear that the senior population has grown enormously across those three decades.

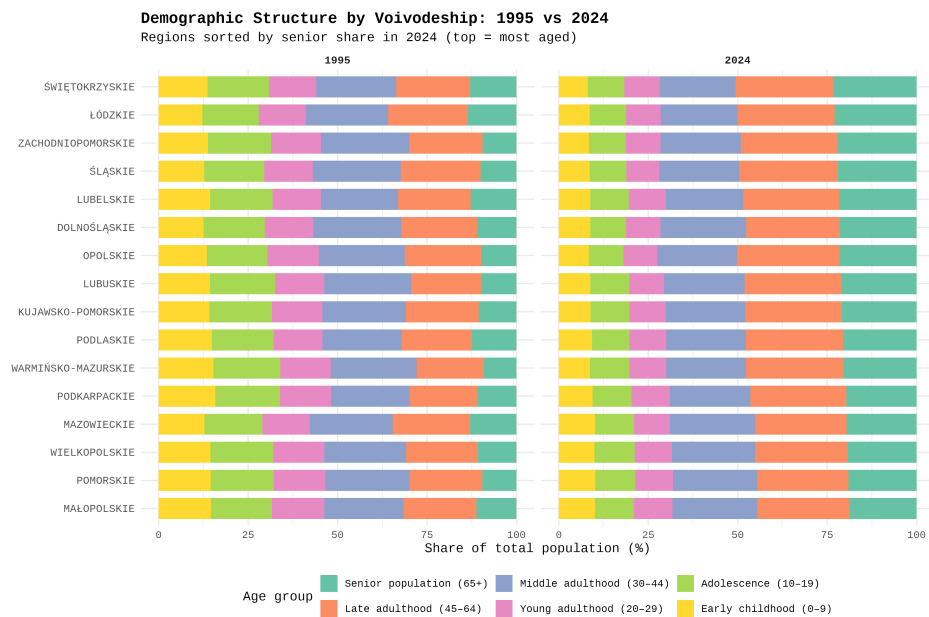


Figure 4: Full demographic structure by voivodeship in 1995 and 2024, sorted by senior share in 2024.

5 Conclusion

The analysis confirms that population ageing is a general trend across all Polish voivodeships between 1995 and 2024. The share of people aged 65+ increased everywhere, while younger age groups, especially children and adolescents, became smaller as a share of the regional population.

The most aged regions in 2024 are Świętokrzyskie, Łódzkie, and Zachodniopomorskie. The relatively youngest regions, using low senior share as the main criterion, are Małopolskie, Pomorskie, and Wielkopolskie. Therefore, the project supports the conclusion that some regions are clearly older or younger than others, while also showing that the differences form a gradual regional ranking rather than isolated extreme cases.